

5(a)

parental phenotypes: ash-red male blue female

parental genotypes: $Z^A Z^{Br} Z^{Bl} W$;

offspring

parental
gametes

		Z^{Bl}		W
Z^A	genotype	$Z^A Z^{Bl}$	genotype	$Z^A W$
	phenotype	ash-red male	phenotype	ash-red female
Z^{Br}	genotype	$Z^{Bl} Z^{Br}$	genotype	$Z^{Br} W$
	phenotype	blue male	phenotype	brown female

*allow one mark for correct parental gametes (in either orientation on the table) ;**allow one mark for correct offspring genotypes ;**allow one mark for correct matching colour **and** sex phenotypes ;**if male parent is incorrect then ecf max 1 for correct offspring genotypes from gametes*

4

5(b)

sex-linkage ;

multiple alleles / dominance hierarchy ;

2

Question	Answer	Marks
5(c)	1 cross (ash-red male) with brown female ; Ignore genotype $Z^{Br} W$ <i>plus any two from:</i> 2 if all offspring are ash-red then (male parent is), $Z^A Z^A / AW$; A homozygous for Z^A 3 if (any) offspring are blue then (male parent is), $Z^A Z^{Bl} / AW$; 4 if (any) offspring are brown then (male parent is), $Z^A Z^{Br} / AW$;	3